

Climate Risk & Housing

GeoMob Oct 2025

Christine Langston

My Background

Education

- City Planning, UCL Bartlett
 - Spatial data science CASA modules
- Engineering, MIT
 - Transportation network modelling
 - Measuring public space interaction

Professional

- PlanX at Open Systems Lab
 - UK Planning Software
- LB of Tower Hamlets
 - GIS Analyst
- Tolemi
 - Mapping software for US cities

Overview

- Research Topic
- Methodology
- Findings
- Adapt to UK
- Takeaways



Heat, drought-parched brush fuel ferocious fire

Official says Northern California blaze is unprecedented this early in the season. He fears 'new normal.'



Climate migration is here. The U.S. must invest accordingly.

By PARAG KHANNA
This week, parts of the Pacific Northwest and British Columbia...

Climate plan tests familiar battle lines

Obama's new rules on emissions would burnish his legacy but also highlight a deep political divide.

By DAVID LAUTER

WASHINGTON — With Monday's release of landmark rules to combat global warming, President Obama is putting into place what probably will be the last piece of his ambitious agenda — one that will test the limits of his authority and the country's deep divisions in the face of a climate crisis.

That division was plain to see in reactions to the new rules, which are intended to change how the nation generates electricity in order to cut emissions of carbon dioxide and other gases blamed for warming the world's climate.

Over the next 15 years, the plan would aim to sharply reduce the use of coal and ramp up the use of wind and solar power. Currently, coal accounts for almost 40% of the nation's electricity, whereas wind and solar power account for 4%. By 2030, if the administration's plan works, renewables would account for 24% of U.S. power generation, edging past coal at 22%.

The plan would boost efforts already underway in coastal states to green up by California, to green up the use of renewable power.

But for those parts of the country still heavily reliant on coal, nearly all of them in the Midwest, Great Plains and South, the rules force a major economic shift that many elected officials have vowed to resist.

(See OASIS, A1)



Flood crisis grips cities with more rain forecast

California cities are in a flood crisis as heavy rain is forecast to hit the state. The crisis is the worst in decades, with cities like San Francisco and San Jose facing major flooding.

Isis suffers worst defeat as Iraqis take Ramadi

ISRAELI
The Islamic State group has suffered its worst defeat yet as Iraqi forces have taken the city of Ramadi from its control.

The INDEPENDENT



The drowning of the North of England

Flash floods hit northern towns, leaving a trail of destruction and loss of life. The flooding was the worst in decades.

The Washington Post

MONDAY, AUGUST 3, 2015

Graham moved up in ranks as reservist

While on Hill, he was promoted twice despite little military activity

By CRAIG WHITLOCK

Of all the candidates vying to become the nation's next commander in chief, none has spent as much time in the military as Sen. Lindsey O. Graham. The summer after a 31-year career in the Air Force, including two decades in the reserve while serving in Congress, Graham is running as a national security hawk, calling himself a "halfhearted leader" with "a lifetime of military service." Images of him wearing camouflage in the field are a common theme in his campaign.

But a detailed examination of Graham's military record — much of it obtained under the Freedom of Information Act — shows that treatment as a reservist was a common theme in his campaign.



Deadly wildfire

Stormy tests for climate proposal

CHANGES REFLECT INTERNAL DEBATE
Energy rule faces legal, political challenges

By JODY WARRICK

Updated Dec 16, 2023 - Economy

America's housing shortage explained in one chart

Mayors to be 'front and centre' of plans to build 1.5 million homes

Deputy Prime Minister Angela Rayner said elected Mayors would be given powers similar to those held by London Mayor Sadiq Khan.

© 17th Dec 2024

Sunday **The Dallas Morning News** Dallas, Texas, Sunday, August 4, 2024

DALLAS POLICE DEPARTMENT

Ex-officer quietly accepts plea deal

SWAT member pleaded guilty to 2020 protest assault; sentenced to probation

By MAGGIE PROSSER and KELLY SMITH

A crowd outside City Hall swelled to several hundred peaceful protesters on May 30, 2020, the second straight day of demonstrations in downtown Dallas demanding police brutality after the murder

NORTH TEXAS

Housing gap dashes dreams



Dania Jones and her husband struggle to pay the bills on a three-bedroom rental home in Little Elm, Jones dreams of having a place to call her own

GRAPHICS: Homeless have not kept pace with home price increases. DFW population outpaces homebuilding. **BA**

Supply crunch puts owning out of reach for many

By NICK WOOTEN

Illustration by @daniellawsonart

Dania Jones' biggest dream is to own a home.

The 42-year-old came to the United States from Honduras as a young girl. She was born in Dallas-Fort Worth from Miami about two decades ago with the promise of a strong economy and cheap housing.

She's come close multiple times in the past few years. But like so many others in DFW, home ownership has eluded her.

Jones said credit issues, a debilitating brain tumor and unforeseen economic circumstances changed her family's life. She's unable to work.

Now, she and her husband, Alan, struggle to pay the bills on a three-bedroom rental

home in Little Elm. Still, she presses on. She wants the freedom to have a nice, clean place to call her own, and she wants something to pass on to her 8-year-old son, Ethan.

"I don't give up on my dream," Jones said. "I know that I'm a believer. I'm a fighter. I'm here for a reason. God gave me the strength to keep fighting."

In Dallas-Fort Worth, housing has been difficult in recent years.

The metro area topped an estimated 8 million residents last year, and the region could pass New York City as the nation's largest metro by the end of the century if migration patterns continue.

There hasn't been enough affordable

See **HOMEBUILDING** Page B4

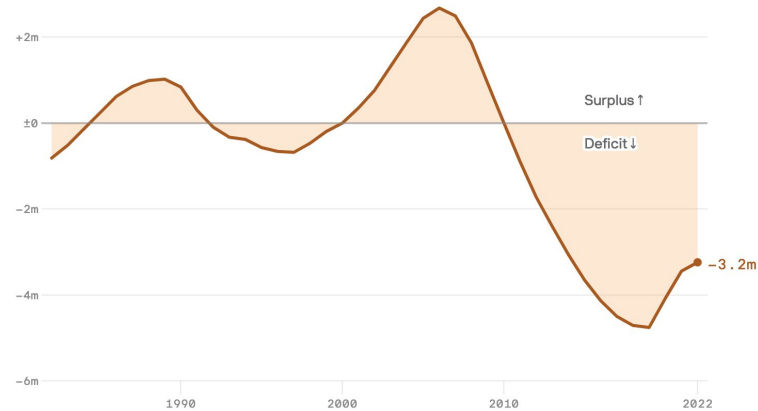
Build 100,000 council houses a year — and make them beautiful, say MPs

A cross-party report has called for an end to the grey, soulless eyesores that came to define many post-war homes. Plus, five council estates that got it right



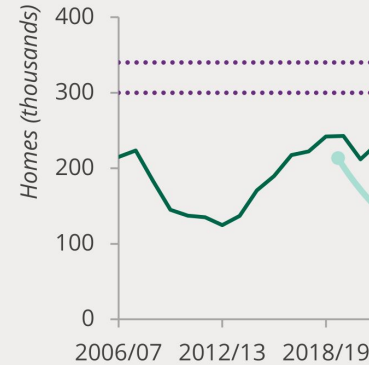
Existing housing units relative to population demand in the U.S.

Annually; 1982-2022



Data: Hines analysis of Census Bureau and Moody's data; Note: Population demand is a theoretical housing demand metric based on long-term household formation and homeownership rates by age cohort; Chart: Axios Visuals

Is supply meeting need?



The Government has said it wants housing supply to reach 300,000 homes a year by the mid-2020s. Others have called for as many as 340,000 a year.

New supply reached a recent peak of 243,000 in 2019/20, but has since fallen.

Source: DLUHC

Explore Climate Risk of New Housing

Climate Risk

- Natural disasters: Floods to wildfires
- Better modeling is helping close the knowledge gap
- Homeowner, Renter, Lender & Insurance implications

Housing Production

- Gap in supply
- Pressure to build quickly
- Infrastructure for future 50-100 years

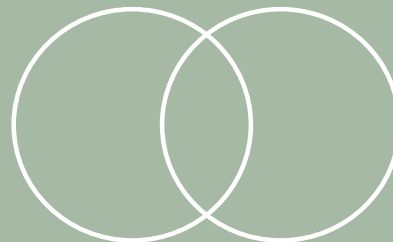
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Reduce Housing in Climate Risk Areas

Aim: protect homes, personal savings, infrastructure investments

- Economic market approach:

The private market (Insurance, Lenders, etc) will make it too costly to build/own/operate in high climate risk areas

- Policy regulation:

Restrict house building to lower risk areas

Findings: new housing has been built at proportionately high rates in higher risk areas. A higher proportion of total building value exists in the higher risk areas compared to the proportion of population in those areas.

Research Methodology

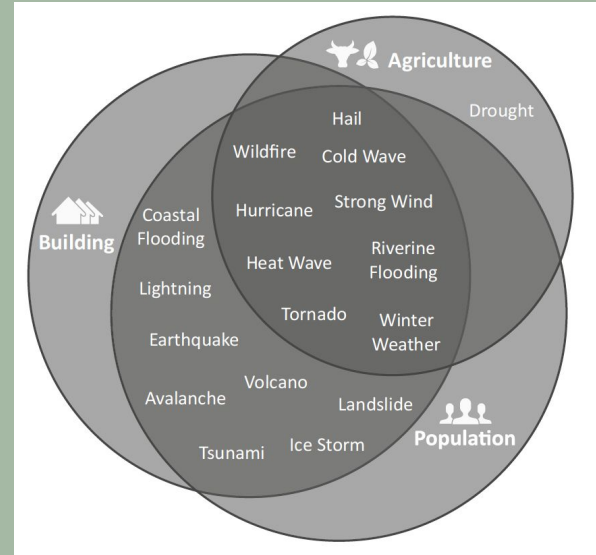
To what extent are *local planning policies* restricting *new housing* to *high climate risk areas*?

Datasets

- Census historic housing production
- Zoning in US - designates location and density
- Federal Govt dataset of 18 natural disasters

Methods

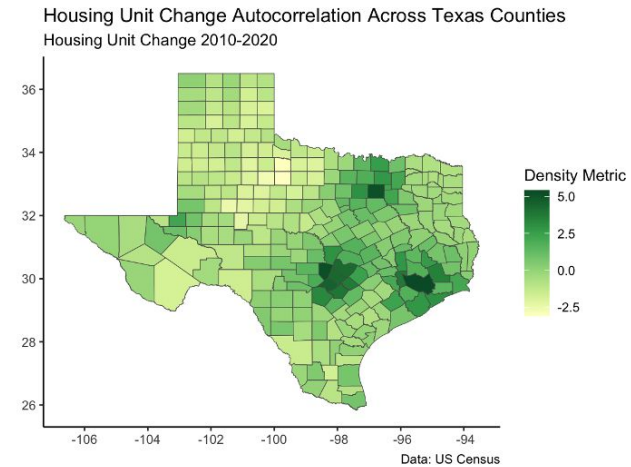
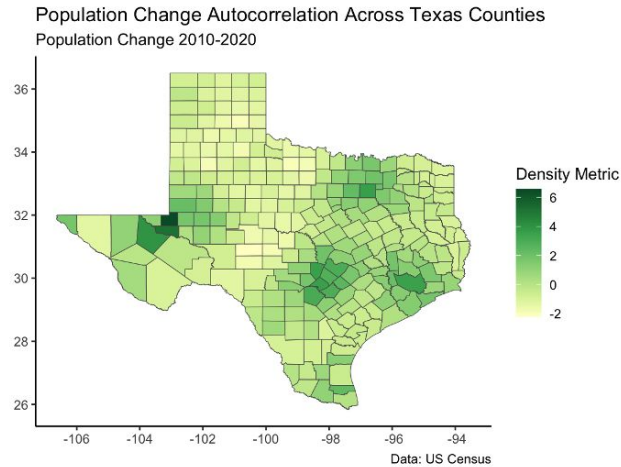
- Autocorrelation
- Spatial intersections



US National Risk Index, FEMA 2023

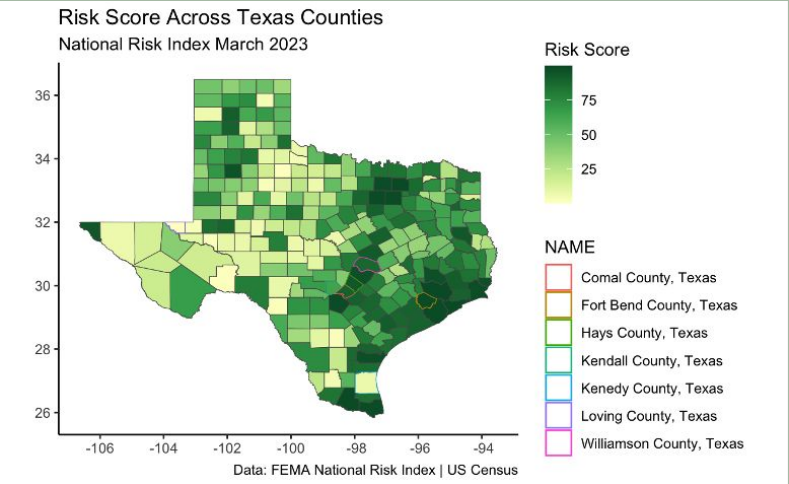
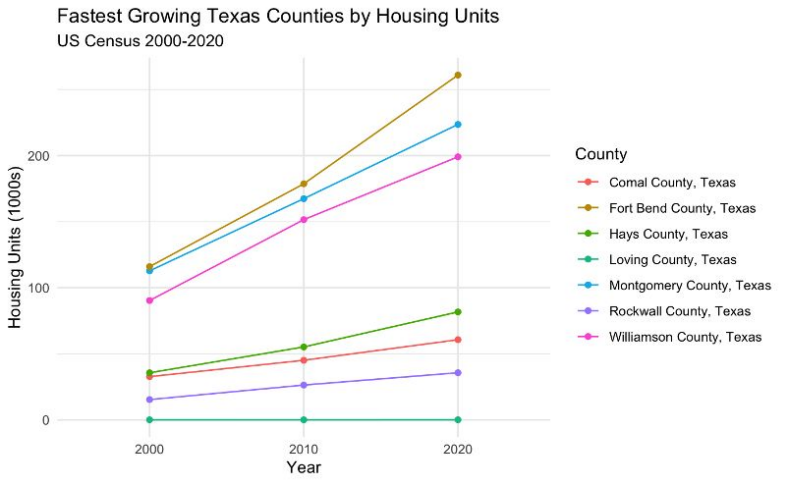
Findings

Population growth and house building clustered in several metropolitan areas in Texas



Findings

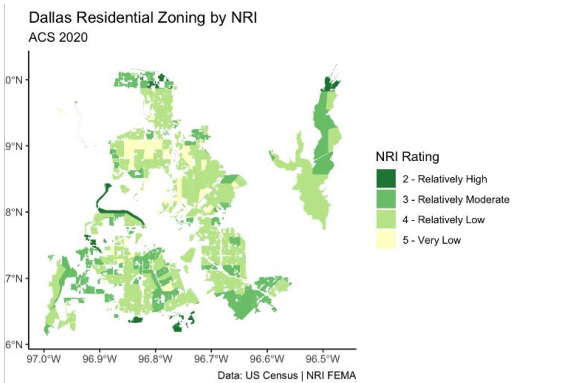
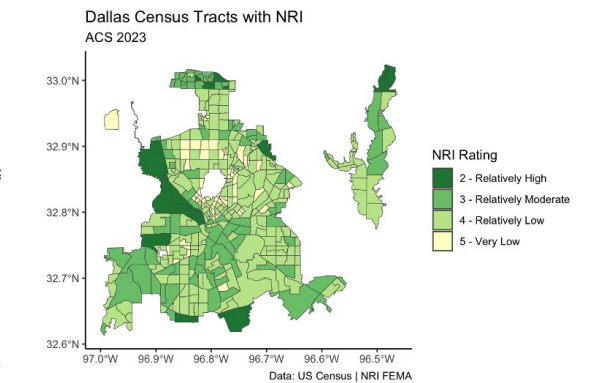
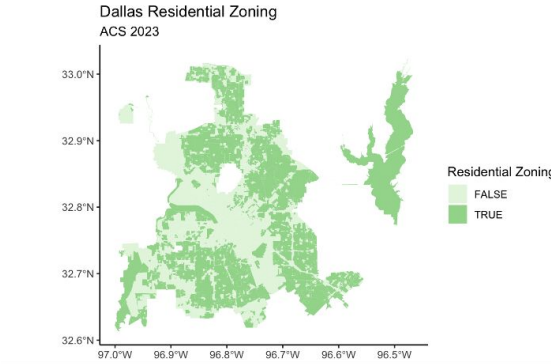
Fast growing counties are in high risk areas



Findings

Dallas zoning policies reserve land for residential building rights that skews towards low-risk areas

	Change from 2010 to 2020
Population	8.90%
Housing	10.18%



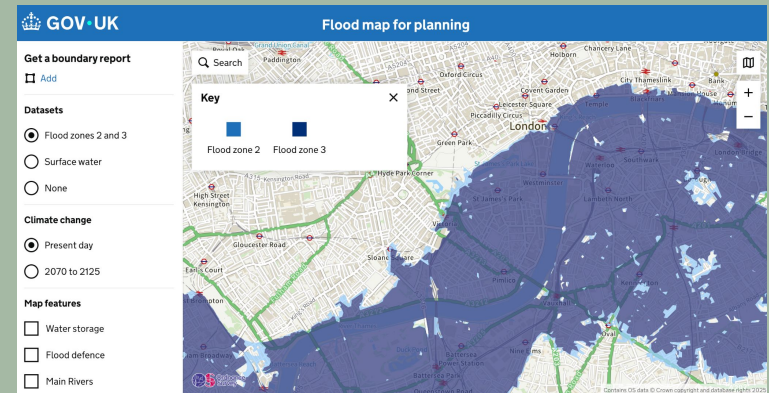
Findings

		Policy		Development			Outcomes	
NRI Rating	Land Area	Residential Zone Land Area	Multi-Unit Zone Land Area	Percent housing 2010	Percent housing 2020	Percent of total growth	Percent Population 2020	Percent Building Value 2023
Relatively High	10.56%	3.67%	2.20%	3.7%	4.2%	8.84%	4.45%	12.26%
Relatively Moderate	31.21%	28.71%	33.70%	28.2%	28.2%	26.66%	28.8%	32.60%
Relatively Low	52.00%	61.26%	58.37%	58.2%	57.1%	45.89%	58.5%	47.28%
Very Low	6.23%	6.36%	5.73%	9.7%	10.5%	18.6%	8.3%	7.86%

Adapt to UK

- Climate Risk
 - Fluvial Flood risk, Coastal flood, Urban Heat Index
- Housing Policy
 - National Planning Policy Framework sets out tests for planning permission
- Site allocations

- Challenges:
 - Planning System optimises for so many factors
 - Where is *low* risk and available land?



Thank you!

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